
BY 511: Cell Physiology in Health and Diseases

Course Code	BY 511
Course Name	Cell Physiology in Health and Diseases
L-T-P-C	3-0-0-3
Prerequisites	IC136, or consent of faculty
Intended for	B. Tech. 3rd and 4th year, MS/M. Tech., Ph.D.
Distribution	Core for M. Tech. Biotechnology, elective for others
Approval	9th Senate (May 2016)

Course Contents

- **Introduction to Cell Physiology**

Introduction to cell physiology, Cells of Eukaryotic system (animal), Cells of Eukaryotic system (Plant)[3 Lectures]

- **Cell membrane physiology I**

Structure and organization of cell membrane and function, Membrane voltage-gated ion channels: molecular structure and regulation of physiological process, Epilepsy and its biology, Alzheimer and its biology, Sick cell anemia, Transport across biological membranes, Active/passive, diffusion, ion channels, aquaporins, Cystic fibrosis, polycystic kidney disease, Deafness, Liddle's syndrome. [12 Lectures]

- **Cell membrane physiology II**

Intra cellular compartments and cell Trafficking, vesicular trafficking, endocytosis and exocytosis, Protein Trafficking, Cell migration and homing, T cell homing. [7 Lectures]

- **Cell Adhesion**

Cell-to-cell signaling: hormones, receptors and intracellular messengers, Adhesion junctions, integrins, Cell adhesion and its role in immune cell maturation and activation, Adhesion and its role in bacterial/viral disease, Cancer initiation and progression, Extracellular matrix and cell-cell interactions. [14 Lectures]

- **Cell Respiration and Free radicals**

Cellular respiration, Generation of free radicals and their role in disease progression [6 Lectures]

Text and Reference Books

1. Nicholas Sperelaki, **Cell Physiology Source Book: A Molecular Approach**, 4th edition.
2. Mordecai P. Blaustein, Mordecai P. Blaustein, Kao Joseph P. Y., Donald R. Matteson, **Cellular Physiology**, 6th edition.
3. Bruce Alberts, Julian Lewis, Alexander Johnson, **Molecular Biology of the Cell**, 6th Edition.